Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/29/2012
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Feasibility Pilot Study on Acute Biological and Behavioral Responses to the Solid State Lighting Modules for the International Space Station		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/15/2009	**End Date:**	02/14/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is now 2/14/2012 per NSSC information (Ed., 9/20/2011) NOTE: End date extended to 8/14/2011 (original end date 8/14/2010), per NSSC info (9/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hanifin, John (Thomas Jefferson University) Lockley, Steven Ph.D. (Harvard Medical School/Brigham and Women's Hospital) Byrne, Brenda Ph.D. (Thomas Jefferson University) Gerner, Edward M.D. (Thomas Jefferson University) Pineda, Carissa M.D. (Thomas Jefferson University) Mollicone, Daniel Ph.D. (Pulsar Informatics) Goel, Namni Ph.D. (University of Pennsylvania)		
Grant/Contract No.:	NNX09AM68G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Currently, the International Space Station (ISS) uses General Luminaire Assemblies (GLAs) that house fluorescent lamps for illuminating the astronauts' working and living environments. Solid-state light in the form of Light Emitting Diodes (LEDs) is an attractive candidate for replacing the GLAs on the ISS as well as all future manned space vehicles and habitats. With respect to spaceflight, the advantages of LEDs over conventional incandescent and fluorescent lighting systems include lower upmass, power consumption, and heat generation, as well as fewer toxic materials. In addition, LEDs have greater resistance to damage and longer lamp life expectancy. Engineers at Kennedy Space Center (KSC) have developed a prototype Solid-State Lighting Assembly (SSLA) that has been installed on board the ISS. In a consensus document between the KSC engineers, lighting experts from Johnson Space Center (JSC), research management from JSC and NASA's biomedical research partner, the National Space Biomedical Research Institute (NSBRI), it was recommended that the SSLAs could provide reasonably equivalent light compared to the GLAs currently onboard ISS. Two crewmembers onboard the ISS have assessed the recently installed SSLA prototype to be acceptable for supporting vision. With regard to the efficacy of the SSLAs for astronaut circadian regulation, there was less certainty. It was suggested that further research was needed to determine the biological and behavioral efficacy of the SSLAs. This project was a feasibility, pilot study. The goal was to develop and run a multiday study with a small cohort of healthy, male, astronaut-aged volunteers. The primary study objective was to determine the best way to test the acute effects of the prototype LED light designed for use on the ISS. Specific measures were taken on visual performance, color discrimination, alertness, and cognitive performance through neurobehavioral testing, measures of plasma melatonin levels, as well as total sleep time, sleep latency, and sleep efficiency using electroencephalography (EEG), electrooculography (EOG), and electrocardiography (EKG).		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The sleep deficits experienced by astronauts during spaceflight can be considered a threat to the success of space missions (Bioastronautics Roadmap, Longnecker and Molins, 2005; NASA Human Research Program Integrated Risk Plan, 2011). The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability, and psychomotor performance (Dijk et al., 2001; Whitmeyer et al., 2010). As a measure to counteract sleep disruptions, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness occurring due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States, nearly 22 million Americans do shift work that interferes with a biologically healthy nocturnal sleep cycle (U.S. Bureau of Labor Statistics, 2007). Shift workers have been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, and cognitive and emotional problems. Development of an in-flight lighting countermeasure that helps resolves circadian and sleep disruption in astronauts is likely to help optimize use of light therapy for patient populations with affective, circadian. and sleep disorders. National Aeronautics and Space Administration (2011) Human Research Program Integrated Research Plan, Revision C, July 2011, HRP-47065, Johnson Space Center, Houston, Texas. Dijk DJ, Neri DF, Wyatt JK, Ronda JM, Riel E, A. R-D, Hughes RJ, Elliott AR, Prisk GK, West JB and Czeisler CA. (2001) Sleep, performance, circadian rhythms, and light-dark cycles during two space shuttle flights. Am J Physiol 281:R1647-R1664. Whitmire AM, Leveton LB, Barger L, Brainard G, Dinges DF, Klerman E and Shea C. (2010) Risk of performance errors due to sleep loss, circadian desynchronization, fatigue, and work overload. In Human Health and Performance Risks of Space Exploration Missions, JC McPhee and JB Charles, eds, pp 85-116, NASA, Johnson Space Center, Houston.		

Task Progress:	This project successfully completed each of the proposed specific aims. Collaborators from four institutions developed and refined controlled, ground-based protocols that assessed the visual, neuroendocrine, alerting, and sleep effects relative to different Correlated Color Temperature (CCT) outputs of the Solid State Lighting Assembly (SSLA). Both the Thomas Jefferson University Institutional Review Board and the NASA Committee for the Protection of Human Subjects approved the study protocols. During the time those activities were completed, two high fidelity simulations of the ISS Crew Sleeping Quarters (CQs) with internal illumination provided by an SSLA were constructed with engineering support from JSC, KSC, and the Lighting Sciences Group. Over 200 astronaut-aged, male volunteers were screened to assure they were physically healthy, psychologically sound, drug-free, and had stable sleep-wake rhythms. Ten subjects were selected from that pool of volunteers to enter the multiday study. Nine of the subjects completed the full multiday study providing data on melatonin secretion, subjective alertness, objective alertness, neurobehavioral responses, and sleep parameters. Eight of those subjects also participated in independent experiments on visual performance and color discrimination inside the CQs. Collateral support from the National Space Biomedical Research Institute made it possible to extend the number of subjects studied, enhance the testing of visual performance and color discrimination, as well as augment analysis of polysomnography for both sleep and alertness measures. As hypothesized, the results of the visual performance and color discrimination testing showed that CCT of 2,300 K and 6,500 K SSLA light supported visual performance as well as indirect midday daylight and common 3,500 K CCT fluorescent light. Also as hypothesized, color vision was significantly compromised under dim 2,300 K SSLA light versus brighter indirect daylight, 3,500 K fluorescent light, and 6,500 K SSLA light. Currently a dim 2,700 K SSLA pre-sleep lighting countermeasure is being considered for use on ISS. If this setting is operationally used, the astronauts should be made aware of the potential color discrimination difficulties and be advised to save any tasks requiring color discrimination for the other lighting conditions. Going forward additional visual testing should be done in high fidelity mock ups of other spaces onboard ISS as they have very different lighting configurations and reflectant surfaces. From the outset, the multiday study was intended as a pilot study to determine the feasibility of using a variety of physiological and behavioral measures within the very confined spaces of the replica ISS CQs. Despite the overall study being insufficiently powered, some statistically significant effects were observed. These included: 1) evening plasma melatonin sample; 2) one evening neurobehavioral test; and 3) one morning neurobehavioral test. The evening melatonin and neurobehavioral results support the hypothesis that evening exposure to dim, long wavelength-shifted SSLA light better prepares an astronaut-aged male subjects for sleep. Further, the morning neurobehavioral results support the hypothesis that bright, short wavelength-shifted SSLA light is more alerting than dimmer, long wavelength-shifted SSLA light. These observations are tempered, however, by the fact that the results may have been due to an order effect not the lighting. A fully powered, crossover study is needed to confirm these observations from the multiday pilot study. As intended, the multiday study provided the ability to determine which dependent variables studied could yield statistically significant results through power analyses. The results of the power analyses varied depending of the study measure being assessed. The number of subjects needed for statistically significant results in future studies will be determined by the dependent variables selected for study. To prepare for manned space missions, NASA utilizes Earth-based research in specific environments with elements that are similar or analogous to some of the conditions of spaceflight. Analog research environments can be in extreme locations such as Antarctica, rugged deserts, the ocean floor, or inside of a volcano. The work reported here represents a middle ground between research done in highly controlled laboratory conditions and NASA's experimental field environments. Finally, the issue of gender needs to be addressed in planning a multiday protocol of seven days or longer. In discussions between the collaborative scientific team and NASA management, it was determined that this feasibility study should include only male subjects to reduce variability across the small sample size. Future studies should include both genders to more accurately reflect the sex distribution in the astronaut corps and future spaceflight crews. NASA's Human Research Program (HRP) Requirements Document, along with the Behavioral Health and Performance section of HRP's Integrated Research Plan (2011), identify the "Risk of performance errors due to sleep loss, circadian desynchronization, fatigue, and work overload" as a major risk for humans during spaceflight missions. What still needs to be addressed are operational lighting protocols for human health and performance during spaceflight. This feasibility study has made important contributions to answering these NASA spaceflight-related needs by providing SSLA and CQ pilot data for power analyses that permit the design of future studies on the efficacy of SSLA lighting. The sleep deficits experienced by astronauts during spaceflight can be considered a threat to the success of space missions. The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability, and psychomotor performance. A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Development of an in-flight lighting countermeasure that helps resolves circadian and sleep disruption in astronauts is likely to help optimize use of light therapy for patient populations with affective, circadian, and sleep disorders.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
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Abstracts for Journals and Proceedings	Brainard GC, Coyle W, Kemp J, West K, Downes M, Ayers M, Warfield B, Hasher D, Whittaker A, Kubey A, Bowen C, Maida J, Lockley S, Goel N, Dinges D, Mollicone D, Hanifin JP. "Solid State Lighting for the International Space Station: Tests of Visual Performance and Melatonin Regulation." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
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